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(57) Abstract :
ABSTRACT OF THE INVENTION 20 Smart food freshness detection system designed to monitor and classify the freshness level of perishable food items in real time. The system employs a combination of gas and temperature sensors integrated with an Arduino microcontroller to analyze environmental parameters associated with spoilage. Traditional methods of freshness inspection often rely on human observation, which can be subjective and inaccurate. The proposed system overcomes these challenges by introducing an automated, sensor-driven approach that measures variations in gas concentration and temperature to determine the freshness condition�categorized as Fresh, Moderate, or Spoiled. The collected data is processed using a calibrated freshness algorithm, ensuring reliable and consistent results. The model can be further enhanced with loT integration for remote monitoring and cloud-based analytics. This innovation provides an efficient, low-cost, and scalable solution for minimizing food waste and ensuring consumer safety through intelligent freshness monitorin

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